Caitlin Roach

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Education

Queen's University — Kingston, ON
Honours Computing + Minor in Mathematics + specialization in machine learning — *Sept. 2024 – present*

Coursework includes Data Structures and Algorithms, Computer Architecture, Discrete Mathematics, Programming Paradigms, Statistics, Vector Calculus, and Software Architecture

Experience

Undergraduate Research Assistant — April 2026 – present
Queen's University — *Kingston, ON*

- Worked with Dr. Erin Meger on pursuit-evasion games on connected graphs
- Built a browser-based graph engine to visualize and solve pursuit-evasion problems
- Wrote a closed-form equation for the cop-barricade number on complete staged bipartite graphs, a novel result

Managing Director of Operations — January 2026 – present
Google Developer Group - QMIND — *Kingston, ON*

- Directed operations for Canada's largest undergraduate AI organization (˜300 members), a chapter of the Google Developer Groups Program
- Maintained QMIND.ca, introducing a central structure via GitHub Organizations
- Managed business banking access and reconciliation for a $77,000 annual operating budget
- Maintained relationships with partners at Google, Deloitte, Microsoft, Amazon, CIBC, and Proxmox

Awards and Honours

Schulich Leadership Award | *Valued at $100,000, awarded by the Schulich Foundation* — May 2024
CSE CTF winner | *Won Communications Security Establishment Canada's CTF challenge.* — February 2026
QHacks: 2nd place | *Made a Chrome extension to detect illegal clauses in ToS.* — January 2025
QHacks: 3rd place | *Used LILYGO Arduino USB Stick as an audio interface for computer systems.* — February 2026

Projects

Raspberry Pi homelab | *Pi 5, Linux, Tailscale, Docker* — September 2026

- Designed and deployed a self-hosted home server running a VPN, media server, and local LLM on a Raspberry Pi 5
- Hardened the server with SSH key-only authentication, a ufw firewall, and automatic security updates, with no ports exposed to the public internet
- Used Ollama and Open WebUI for user-friendly access to llama3.2:3b

Adversarial Network Simulator | *Python, JavaScript, Git* — April 2026

- Built an interactive browser simulation of an adversarial pursuit-evasion game, where defenders try to trap an attacker who can block paths
- Developed a Python game engine and graph builder that generates network structures and exports them to JSON for visualization
- Proved bounds on the number of defenders needed to guarantee capture on multistage networks (undergrad research with Dr. Meger)

Technical Skills

Languages: Python, Java, JS/CSS/HTML, C, Shell, x86asm, a little COBOL
Frameworks: React, Node.js, Vite, Pytorch/TensorFlow, MediaPipe
Tools: Git, Linux, Burp Suite, nmap, WireShark
Libraries: pandas, NumPy, Matplotlib